

1. Define Cloud Computing and explain any four characteristics that make it different from traditional on-premise computing.

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, software & analytics over the Internet on a pay-as-you-go basis.

Characteristics:

1. On-Demand self-Service: Users can access computing resources whenever required without human interaction with the service provider.
2. Broad Network Access: Cloud services are available over the Internet and can be accessed from laptops, smartphones and tablets.
3. Resource Pooling: Multiple users share the same infrastructure securely using virtualization.
4. Rapid Elasticity: Resources can be increased or decreased quickly based on user demand.

Difference from Traditional Computing:

Traditional on-premise systems require organizations to purchase, install and maintain hardware, whereas cloud computing provides scalable resources over the Internet with lower maintenance costs.

2. Trace the evolution of Cloud Computing from hardware evolution to Internet software evolution.

1. Hardware Evolution: Large mainframe computers were used to perform centralized computing.

2. Personal Computing: Affordable Personal-Computers enabled individual computing.

3. Network Era: Computers were connected through LANs for resource sharing.

4. Internet Evolution: The Internet enabled communication and information sharing across the world.

5. Internet Software Evolution: Web-based applications & virtualization Technologies allowed software and infrastructure to be delivered as cloud services, leading to modern cloud platforms such as AWS, Microsoft Azure and Google Cloud.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization and cloud computing.

Role of Server Virtualization:

→ It divides one physical server into multiple virtual machines.

→ Each virtual machine works independently with its own operating system.

→ It Improves hardware utilization and reduces costs.

→ It enables scalability and efficient resource management.

Difference:

Virtualization	Cloud Computing.
Technology That creates virtual machines	Service models that deliver computing resources over the Internet.
Focuses on efficient hardware utilization.	Focuses on providing services on demand.
Forms the foundation of Cloud Computing	Uses virtualization to deliver services like IaaS, PaaS and SaaS

4. Discuss The role of data centres in delivering cloud services.

A data centre is a facility that contains servers, storage devices, networking equipment, and power systems used to deliver cloud services.

Roles:

→ Stores and processes user data.

→ Hosts cloud applications and websites.

→ Provides high availability through backup and disaster recovery.

→ Ensures security using fire walls, encryptions, and access control.

→ Enables scalability by adding or removing computing resources based on demand.

Thus, data centres are the backbone of cloud computing because they provide reliable, secure and continuous cloud services.

5. Differentiate IaaS, PaaS, SaaS & XaaS with one real world example for each.

Service Model	Description	Example
IaaS (Infrastructure as a Service)	Provides virtual servers, storage & networking. Users manage the OS and application.	Amazon EC2
PaaS (Platform as a Service)	Provides a platform to develop, test and deploy applications without managing hardware	Google App Engine.
SaaS (Software as a Service)	Delivers Software applications through a web browser.	Google Workspace or Microsoft 365.
XaaS (Anything as a Service)	Offers any IT service through the cloud, such as databases, AI or security services	Microsoft Azure AI Services.

These service models provide increasing levels of abstraction, making cloud computing flexible, scalable and cost-effective.